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Studierichting: Informatica
Oefeningenreeks 4A nr. 43.8

$$\int \frac{2x^2+3x+3}{(x+1)^3}dx$$

$$\frac{2x^2+3x+3}{(x+1)^3}dx = \frac{A}{(x+1)^3} + \frac{B}{(x+1)^2} + \frac{C}{x+1}$$

$$A = 2x^2+3x+3\big|_{x=-1} = 2$$

$$\frac{2x^2+3x+3}{(x+1)^3} - \frac{2}{(x+1)^3} = \frac{2x^2+3x+2}{(x+1)^3} = \frac{(x+1)(2x+1)}{(x+1)^3} = \frac{2x+1}{(x+1)^2}$$

$$B = 2x+1\big|_{x=-1} = -1$$

$$\frac{2x+1}{(x+1)^2} - \frac{-1}{(x+1)^2} = \frac{2(x+1)}{(x+1)^2} = \frac{2}{x+1}$$

$$C = 2\big|_{x=-1} = 2$$

$$\int \frac{2x^2+3x+3}{(x+1)^3}dx = \int \frac{2dx}{(x+1)^3} - \int \frac{dx}{(x+1)^2} + \int \frac{2dx}{x+1}$$

Stel $t = x+1 \Rightarrow dt = dx$

$$= 2\int \frac{dt}{t^3} - \int \frac{dt}{t^2} + 2\int \frac{dt}{t}$$

$$= 2\frac{t^{-2}}{-2} - \frac{t^{-1}}{-1} + 2\ln|t| + C$$

$$= \frac{-1}{t^2} + \frac{1}{t} + 2\ln|t| + C$$

$$= \frac{-1}{(x+1)^2} + \frac{1}{x+1} + 2\ln|x+1| + C$$

References